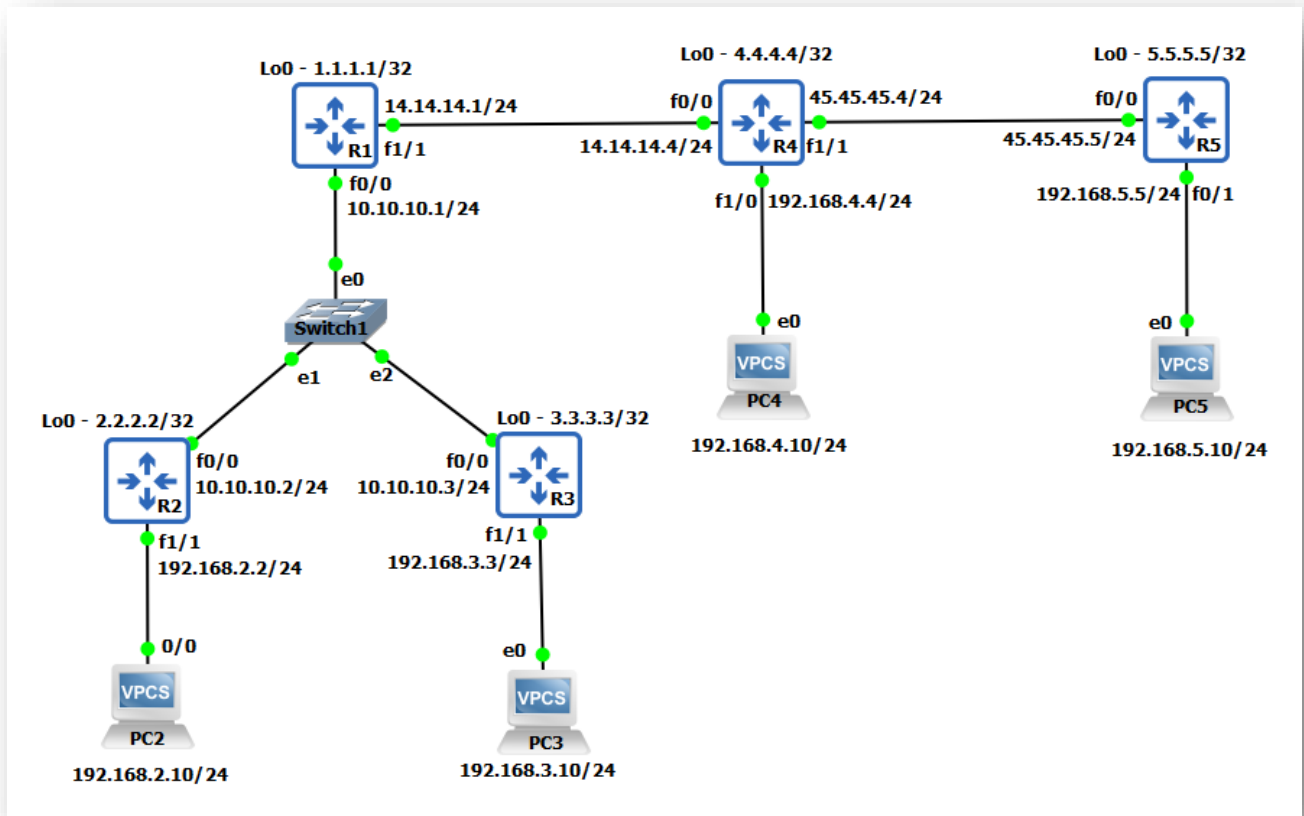


OSPFv2 – Single Area



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** & device access password as **cisco**
2. Configure OSPF process for single area-0 (backbone) on all routers and advertise its directly connected networks.
3. All router should be able to ping each other's Loopback IP.
4. PCs should be able to ping each other's IP.
5. Configure clear text authentication between R1 & R4.
6. Configure MD5 authentication between R4 & R5
7. Modify the Hello and Dead Interval between R1 & R4

Task-1: Routers Configuration:

R1 Router Configuration:

```

R1#enable
R1#config t
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#int fa0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0

```

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```
R1(config-if)#description Link to Switch-1
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int fa1/1
R1(config-if)#ip address 14.14.14.1 255.255.255.0
R1(config-if)#description Link to R2 on fa0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int lo0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#
```

R2 Router Configuration:

```
R2#enable
R2#config t
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#int fa0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#description Link to Switch-1
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa1/1
R2(config-if)#ip address 192.168.2.2 255.255.255.0
R2(config-if)#description Link to local LAN
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int lo0
R2(config-if)#ip address 2.2.2.2 255.255.255.0
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0
```

```
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#
```

R3 Router Configuration:

```
R3#enable
R3#config t
R3(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#int fa0/0
R3(config-if)#ip address 10.10.10.3 255.255.255.0
R3(config-if)#description Link to Switch-1
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa1/1
R3(config-if)#ip address 192.168.3.3 255.255.255.0
R3(config-if)#description Link to local LAN
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int lo0
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#
```

R4 Router Configuration:

```
R4#enable
R4(config)#hostname R4
R4(config)#no ip domain-lookup
R4(config)#enable password ccna
```

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```
R4(config)#int fa0/0
R4(config-if)#ip address 14.14.14.4 255.255.255.0
R4(config-if)#description Link to R1 on fa1/1
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa1/0
R4(config-if)#ip address 192.168.4.4 255.255.255.0
R4(config-if)#description Link to local LAN
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa1/1
R4(config-if)#ip address 45.45.45.4 255.255.255.0
R4(config-if)#description Link to R5 on fa0/0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int lo0
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#exit
R4(config)#line vty 0 15
R4(config-line)#password cisco
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 1 0
R4(config-line)#exit
R4(config)#line console 0
R4(config-line)#password cisco1
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 0 0
R4(config-line)#exit
R4(config)#exit
R4#
```

R5 Router Configuration:

```
R5#enable
R5#config t
R5(config)#hostname R5
R5(config)#no ip domain-lookup
R5(config)#enable password ccna
R5(config)#int fa0/0
R5(config-if)#ip address 45.45.45.5 255.255.255.0
R5(config-if)#description Link to R4 on fa1/1
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int fa0/1
R5(config-if)#ip address 192.168.5.5 255.255.255.0
R5(config-if)#description Link to local LAN
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int lo0
```

```

R5(config-if)#ip address 5.5.5.5 255.255.255.0
R5(config-if)#exit
R5(config)#line vty 0 15
R5(config-line)#password cisco
R5(config-line)#login
R5(config-line)#logging synchronous
R5(config-line)#exec-timeout 1 0
R5(config-line)#exit
R5(config)#line console 0
R5(config-line)#password cisco1
R5(config-line)#login
R5(config-line)#logging synchronous
R5(config-line)#exec-timeout 0 0
R5(config-line)#exit
R5(config)#exit
R5#

```

PC2 Configuration:

```

PC2> ip 192.168.2.10/24 192.168.2.2
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.2

```

PC3 Configuration:

```

PC3> ip 192.168.3.10/24 192.168.3.3
Checking for duplicate address...
PC1 : 192.168.3.10 255.255.255.0 gateway 192.168.3.3

```

PC4 Configuration:

```

PC4> ip 192.168.4.10/24 192.168.4.4
Checking for duplicate address...
PC1 : 192.168.4.10 255.255.255.0 gateway 192.168.4.4

```

PC5 Configuration:

```

PC5> ip 192.168.5.10/24 192.168.5.5
Checking for duplicate address...
PC1 : 192.168.5.10 255.255.255.0 gateway 192.168.5.5

```

✓ Verification & Testing:**R1#sh ip int brief | exclude unassign**

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.1	YES	manual	up	up
FastEthernet1/1	14.14.14.1	YES	manual	up	up
Loopback0	1.1.1.1	YES	manual	up	up

R1#

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R2#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.2	YES	manual	up	up
FastEthernet1/1	192.168.2.2	YES	manual	up	up
Loopback0	2.2.2.2	YES	manual	up	up

R2#

R3#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.3	YES	manual	up	up
FastEthernet1/1	192.168.3.3	YES	manual	up	up
Loopback0	3.3.3.3	YES	manual	up	up

R3#

R4#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	14.14.14.4	YES	manual	up	up
FastEthernet1/0	192.168.4.4	YES	manual	up	up
FastEthernet1/1	45.45.45.4	YES	manual	up	up
Loopback0	4.4.4.4	YES	manual	up	up

R4#

R5#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	45.45.45.5	YES	manual	up	up
FastEthernet0/1	192.168.5.5	YES	manual	up	up
Loopback0	5.5.5.5	YES	manual	up	up

R5#

Task -2: Configure OSPF process for single area - 0 (backbone) on all routers and advertise its directly connected networks.

R1 OSPF Configuration:

```
R1#config t
R1(config)#router ospf 1
R1(config-router)#network 14.14.14.1 0.0.0.0 area 0
R1(config-router)#network 10.10.10.0 0.0.0.255 area 0
R1(config-router)#network 1.1.1.1 0.0.0.0 area 0
R1(config-router)#exit
R1(config)#exit
R1#
```

R2 OSPF Configuration:

```
R2#config t
R2(config)#router ospf 2
R2(config-router)#network 192.168.2.0 0.0.0.255 area 0
R2(config-router)#network 10.10.10.0 0.0.0.255 area 0
R2(config-router)#network 2.2.2.2 0.0.0.0 area 0
R2(config-router)#exit
```

```
R2(config)#exit
R2#
```

R3 OSPF Configuration:

```
R3#config t
R3(config)#router ospf 3
R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#network 10.10.10.0 0.0.0.255 area 0
R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#exit
R3(config)#exit
R3#
```

R4 OSPF Configuration:

```
R4#config t
R4(config)#router ospf 4
R4(config-router)#network 192.168.4.0 0.0.0.255 area 0
R4(config-router)#network 14.14.14.4 0.0.0.0 area 0
R4(config-router)#network 45.45.45.4 0.0.0.0 area 0
R4(config-router)#network 4.4.4.4 0.0.0.0 area 0
R4(config-router)#exit
R4(config)#exit
R4#
```

R5 OSPF Configuration:

```
R5#config t
R5(config)#int fa0/0
R5(config-if)#ip ospf 5 area 0
R5(config-if)#exit
R5(config)#int fa0/1
R5(config-if)#ip ospf 5 area 0
R5(config-if)#exit
R5(config)#int lo0
R5(config-if)#ip ospf 5 area 0
R5(config-if)#exit
R5(config)#exit
R5#
```

OR

```
R5#config t
R5(config)#router ospf 5
R5(config-router)#network 192.168.5.0 0.0.0.255 area 0
R5(config-router)#network 45.45.45.5 0.0.0.0 area 0
R5(config-router)#network 5.5.5.5 0.0.0.0 area 0
R5(config-router)#exit
R5(config)#exit
R5#
```

✓ **Verification & Testing:****R1#sh ip protocols**

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 1.1.1.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

1.1.1.1 0.0.0.0 area 0

10.10.10.0 0.0.0.255 area 0

14.14.14.1 0.0.0.0 area 0

Routing Information Sources:

Gateway	Distance	Last Update
5.5.5.5	110	00:01:42
4.4.4.4	110	00:03:13
3.3.3.3	110	00:10:54
2.2.2.2	110	00:11:13

Distance: (default is 110)**R2#sh ip protocols**

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 2"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 2.2.2.2

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

2.2.2.2 0.0.0.0 area 0

10.10.10.0 0.0.0.255 area 0

192.168.2.0 0.0.0.255 area 0

Routing Information Sources:

Gateway	Distance	Last Update
5.5.5.5	110	00:01:51
4.4.4.4	110	00:03:22
3.3.3.3	110	00:10:57
1.1.1.1	110	00:11:26

Distance: (default is 110)**R3#sh ip protocols**

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 3"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 3.3.3.3

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

3.3.3.3 0.0.0.0 area 0

10.10.10.0 0.0.0.255 area 0

192.168.3.0 0.0.0.255 area 0

Routing Information Sources:

Gateway	Distance	Last Update
5.5.5.5	110	00:01:53
4.4.4.4	110	00:03:25
1.1.1.1	110	00:11:02
2.2.2.2	110	00:11:02

Distance: (default is 110)

R4#sh ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 4"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 4.4.4.4

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

4.4.4.4 0.0.0.0 area 0

14.14.14.4 0.0.0.0 area 0

45.45.45.4 0.0.0.0 area 0

192.168.4.0 0.0.0.255 area 0

Routing Information Sources:

Gateway	Distance	Last Update
5.5.5.5	110	00:02:00
1.1.1.1	110	00:10:29
2.2.2.2	110	00:10:29
3.3.3.3	110	00:10:29

Distance: (default is 110)

R5#sh ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 5"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 5.5.5.5

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

Routing on Interfaces Configured Explicitly (Area 0):

Loopback0

FastEthernet0/1

FastEthernet0/0

Routing Information Sources:

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Gateway	Distance	Last Update
1.1.1.1	110	00:02:03
2.2.2.2	110	00:02:03
3.3.3.3	110	00:02:03
4.4.4.4	110	00:02:03

Distance: (default is 110)

OSPF Default Cost

Interface type	bandwidth	Cost
Fast Ethernet and faster	100 Mb/s and higher	1
Ethernet	10 Mb/s	10
E1	2 Mb/s	48
T1	1.544 Mb/s	64
128bps	128bps	781
64kbps	64kbps	1562
56kbps	56kbps	1785

R1#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	1	0	1.1.1.1/24	1	LOOP	0/0
Fa0/0	1	0	10.10.10.1/24	1	DR	2/2
Fa1/1	1	0	14.14.14.1/24	1	DR	1/1

R1#

R2#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	2	0	2.2.2.2/24	1	LOOP	0/0
Fa0/0	2	0	10.10.10.2/24	1	BDR	2/2
Fa1/1	2	0	192.168.2.2/24	1	DR	0/0

R2#

R3#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	3	0	3.3.3.3/24	1	LOOP	0/0
Fa0/0	3	0	10.10.10.3/24	1	DROTH	2/2
Fa1/1	3	0	192.168.3.3/24	1	DR	0/0

R3#

R4#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	4	0	4.4.4.4/24	1	LOOP	0/0
Fa1/1	4	0	45.45.45.4/24	1	DR	1/1
Fa0/0	4	0	14.14.14.4/24	1	BDR	1/1
Fa1/0	4	0	192.168.4.4/24	1	DR	0/0

R4#

R5#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	5	0	5.5.5.5/24	1	LOOP	0/0
Fa0/1	5	0	192.168.5.5/24	1	DR	0/0

Fa0/0 5 0 45.45.45.5/24 1 BDR 1/1
R5#

R1#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/2] via 10.10.10.2, 00:14:48, FastEthernet0/0
3.0.0.0/32 is subnetted, 1 subnets
O 3.3.3.3 [110/2] via 10.10.10.3, 00:14:29, FastEthernet0/0
4.0.0.0/32 is subnetted, 1 subnets
O 4.4.4.4 [110/2] via 14.14.14.4, 00:13:49, FastEthernet1/1
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/3] via 14.14.14.4, 00:05:17, FastEthernet1/1
45.0.0.0/24 is subnetted, 1 subnets
O 45.45.45.0 [110/2] via 14.14.14.4, 00:06:48, FastEthernet1/1
O 192.168.2.0/24 [110/2] via 10.10.10.2, 00:14:48, FastEthernet0/0
O 192.168.3.0/24 [110/2] via 10.10.10.3, 00:14:29, FastEthernet0/0
O 192.168.4.0/24 [110/2] via 14.14.14.4, 00:13:49, FastEthernet1/1
O 192.168.5.0/24 [110/3] via 14.14.14.4, 00:05:17, FastEthernet1/1

R2#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/2] via 10.10.10.1, 00:14:57, FastEthernet0/0
3.0.0.0/32 is subnetted, 1 subnets
O 3.3.3.3 [110/2] via 10.10.10.3, 00:14:28, FastEthernet0/0
4.0.0.0/32 is subnetted, 1 subnets
O 4.4.4.4 [110/3] via 10.10.10.1, 00:13:44, FastEthernet0/0
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/4] via 10.10.10.1, 00:05:22, FastEthernet0/0
14.0.0.0/24 is subnetted, 1 subnets

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```
O 14.14.14.0 [110/2] via 10.10.10.1, 00:14:57, FastEthernet0/0
45.0.0.0/24 is subnetted, 1 subnets
O 45.45.45.0 [110/3] via 10.10.10.1, 00:06:53, FastEthernet0/0
O 192.168.3.0/24 [110/2] via 10.10.10.3, 00:14:28, FastEthernet0/0
O 192.168.4.0/24 [110/3] via 10.10.10.1, 00:13:44, FastEthernet0/0
O 192.168.5.0/24 [110/4] via 10.10.10.1, 00:05:22, FastEthernet0/0
R2#
```

R3#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, **O - OSPF**, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/2] via 10.10.10.1, 00:14:34, FastEthernet0/0
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/2] via 10.10.10.2, 00:14:34, FastEthernet0/0
4.0.0.0/32 is subnetted, 1 subnets
O 4.4.4.4 [110/3] via 10.10.10.1, 00:13:48, FastEthernet0/0
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/4] via 10.10.10.1, 00:05:26, FastEthernet0/0
14.0.0.0/24 is subnetted, 1 subnets
O 14.14.14.0 [110/2] via 10.10.10.1, 00:14:34, FastEthernet0/0
45.0.0.0/24 is subnetted, 1 subnets
O 45.45.45.0 [110/3] via 10.10.10.1, 00:06:57, FastEthernet0/0
O 192.168.2.0/24 [110/2] via 10.10.10.2, 00:14:34, FastEthernet0/0
O 192.168.4.0/24 [110/3] via 10.10.10.1, 00:13:48, FastEthernet0/0
O 192.168.5.0/24 [110/4] via 10.10.10.1, 00:05:26, FastEthernet0/0
R3#
```

R4#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, **O - OSPF**, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/2] via 14.14.14.1, 00:14:00, FastEthernet0/0
2.0.0.0/32 is subnetted, 1 subnets
```

```

O    2.2.2.2 [110/3] via 14.14.14.1, 00:14:00, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O    3.3.3.3 [110/3] via 14.14.14.1, 00:14:00, FastEthernet0/0
    5.0.0.0/32 is subnetted, 1 subnets
O    5.5.5.5 [110/2] via 45.45.45.5, 00:05:31, FastEthernet1/1
    10.0.0.0/24 is subnetted, 1 subnets
O    10.10.10.0 [110/2] via 14.14.14.1, 00:14:00, FastEthernet0/0
O    192.168.2.0/24 [110/3] via 14.14.14.1, 00:14:00, FastEthernet0/0
O    192.168.3.0/24 [110/3] via 14.14.14.1, 00:14:00, FastEthernet0/0
O    192.168.5.0/24 [110/2] via 45.45.45.5, 00:05:31, FastEthernet1/1
R4#

```

R5#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```

    1.0.0.0/32 is subnetted, 1 subnets
O    1.1.1.1 [110/3] via 45.45.45.4, 00:05:37, FastEthernet0/0
    2.0.0.0/32 is subnetted, 1 subnets
O    2.2.2.2 [110/4] via 45.45.45.4, 00:05:37, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O    3.3.3.3 [110/4] via 45.45.45.4, 00:05:37, FastEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O    4.4.4.4 [110/2] via 45.45.45.4, 00:05:37, FastEthernet0/0
    10.0.0.0/24 is subnetted, 1 subnets
O    10.10.10.0 [110/3] via 45.45.45.4, 00:05:37, FastEthernet0/0
    14.0.0.0/24 is subnetted, 1 subnets
O    14.14.14.0 [110/2] via 45.45.45.4, 00:05:37, FastEthernet0/0
O    192.168.2.0/24 [110/4] via 45.45.45.4, 00:05:37, FastEthernet0/0
O    192.168.3.0/24 [110/4] via 45.45.45.4, 00:05:37, FastEthernet0/0
O    192.168.4.0/24 [110/2] via 45.45.45.4, 00:05:37, FastEthernet0/0
R5#

```

R1#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:32	10.10.10.2	FastEthernet0/0
3.3.3.3	1	FULL/DROTHER	00:00:36	10.10.10.3	FastEthernet0/0
4.4.4.4	1	FULL/BDR	00:00:34	14.14.14.4	FastEthernet1/1

R1#

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R2#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:37	10.10.10.1	FastEthernet0/0
3.3.3.3	1	FULL/DROTHER	00:00:34	10.10.10.3	FastEthernet0/0

R2#

R3#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:34	10.10.10.1	FastEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:36	10.10.10.2	FastEthernet0/0

R3#

R4#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/BDR	00:00:37	45.45.45.5	FastEthernet1/1
1.1.1.1	1	FULL/DR	00:00:31	14.14.14.1	FastEthernet0/0

R4#

R5#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
4.4.4.4	1	FULL/DR	00:00:37	45.45.45.4	FastEthernet0/0

R5#

R1#sh ip ospf database

OSPF Router with ID (1.1.1.1) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1160	0x80000003	0x00AEC0	3
2.2.2.2	2.2.2.2	1220	0x80000002	0x00FA52	3
3.3.3.3	3.3.3.3	1198	0x80000003	0x000A34	3
4.4.4.4	4.4.4.4	650	0x80000005	0x004DA1	4
5.5.5.5	5.5.5.5	651	0x80000001	0x0078D5	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.10.1	1.1.1.1	1201	0x80000002	0x006198
14.14.14.1	1.1.1.1	1160	0x80000001	0x00688E
45.45.45.4	4.4.4.4	650	0x80000001	0x002852

R1#

R2#sh ip ospf database

OSPF Router with ID (2.2.2.2) (Process ID 2)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1298	0x80000003	0x00AEC0	3
2.2.2.2	2.2.2.2	1355	0x80000002	0x00FA52	3
3.3.3.3	3.3.3.3	1334	0x80000003	0x000A34	3
4.4.4.4	4.4.4.4	787	0x80000005	0x004DA1	4
5.5.5.5	5.5.5.5	788	0x80000001	0x0078D5	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.10.1	1.1.1.1	1338	0x80000002	0x006198
14.14.14.1	1.1.1.1	1298	0x80000001	0x00688E
45.45.45.4	4.4.4.4	787	0x80000001	0x002852

R2#

R3#sh ip ospf database

OSPF Router with ID (3.3.3.3) (Process ID 3)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1300	0x80000003	0x00AEC0	3
2.2.2.2	2.2.2.2	1358	0x80000002	0x00FA52	3
3.3.3.3	3.3.3.3	1336	0x80000003	0x000A34	3
4.4.4.4	4.4.4.4	790	0x80000005	0x004DA1	4
5.5.5.5	5.5.5.5	791	0x80000001	0x0078D5	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.10.1	1.1.1.1	1340	0x80000002	0x006198
14.14.14.1	1.1.1.1	1300	0x80000001	0x00688E
45.45.45.4	4.4.4.4	790	0x80000001	0x002852

R3#

R4#sh ip ospf database

OSPF Router with ID (4.4.4.4) (Process ID 4)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1302	0x80000003	0x00AEC0	3
2.2.2.2	2.2.2.2	1361	0x80000002	0x00FA52	3
3.3.3.3	3.3.3.3	1340	0x80000003	0x000A34	3
4.4.4.4	4.4.4.4	790	0x80000005	0x004DA1	4
5.5.5.5	5.5.5.5	791	0x80000001	0x0078D5	3

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Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.10.1	1.1.1.1	1342	0x80000002	0x006198
14.14.14.1	1.1.1.1	1302	0x80000001	0x00688E
45.45.45.4	4.4.4.4	790	0x80000001	0x002852

R4#

R5#sh ip ospf database

OSPF Router with ID (5.5.5.5) (Process ID 5)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1306	0x80000003	0x00AEC0	3
2.2.2.2	2.2.2.2	1364	0x80000002	0x00FA52	3
3.3.3.3	3.3.3.3	1343	0x80000003	0x000A34	3
4.4.4.4	4.4.4.4	793	0x80000005	0x004DA1	4
5.5.5.5	5.5.5.5	792	0x80000001	0x0078D5	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.10.1	1.1.1.1	1345	0x80000002	0x006198
14.14.14.1	1.1.1.1	1306	0x80000001	0x00688E
45.45.45.4	4.4.4.4	793	0x80000001	0x002852

R5#

Ping Loopback IP of all other routers from R1

R1#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/17/24 ms

R1#ping 3.3.3.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/15/20 ms

R1#ping 4.4.4.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/20 ms

R1#ping 5.5.5.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/41/44 ms

Ping Loopback IP of all other routers from R2

R2#ping 1.1.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/24 ms

R2#ping 3.3.3.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/20 ms

R2#ping 4.4.4.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/41/44 ms

R2#ping 5.5.5.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/57/64 ms

Ping Loopback IP of all other routers from R3

R3#ping 1.1.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/21/28 ms

R3#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/20 ms

R3#ping 4.4.4.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 32/41/44 ms

R3#ping 5.5.5.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/58/64 ms

Ping Loopback IP of all other routers from R4**R4#ping 1.1.1.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/15/20 ms

R4#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/41/44 ms

R4#ping 3.3.3.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/41/60 ms

R4#ping 5.5.5.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/20 ms

Ping Loopback IP of all other routers from R5**R5#ping 1.1.1.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/41/64 ms

R5#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/58/64 ms

R5#ping 3.3.3.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/57/64 ms

R5#ping 4.4.4.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/20 ms

Ping all other PCs from PC2**PC2> ping 192.168.3.10**

84 bytes from 192.168.3.10 icmp_seq=1 ttl=62 time=63.226 ms

84 bytes from 192.168.3.10 icmp_seq=2 ttl=62 time=42.221 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=33.286 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=37.169 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=43.202 ms

PC2> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=61 time=118.753 ms

84 bytes from 192.168.4.10 icmp_seq=2 ttl=61 time=59.219 ms

84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=65.127 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=65.089 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=61.155 ms

PC2> ping 192.168.5.10

192.168.5.10 icmp_seq=1 timeout

192.168.5.10 icmp_seq=2 timeout

84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=86.969 ms

84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=84.068 ms

84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=78.097 ms

Ping all other PCs from PC3**PC3> ping 192.168.2.10**

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=42.073 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=35.206 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=32.235 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=41.414 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=34.192 ms

PC3> ping 192.168.4.10

192.168.4.10 icmp_seq=1 timeout

192.168.4.10 icmp_seq=2 timeout

84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=60.108 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=58.041 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=61.997 ms

PC3> ping 192.168.5.10

192.168.5.10 icmp_seq=1 timeout

192.168.5.10 icmp_seq=2 timeout

84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=87.980 ms

84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=80.901 ms

84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=79.922 ms

Ping all other PCs from PC4**PC4> ping 192.168.2.10**

192.168.2.10 icmp_seq=1 timeout

192.168.2.10 icmp_seq=2 timeout

84 bytes from 192.168.2.10 icmp_seq=3 ttl=61 time=62.069 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=61 time=61.953 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=61 time=64.037 ms

PC4> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=61 time=59.480 ms

84 bytes from 192.168.3.10 icmp_seq=2 ttl=61 time=61.908 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=56.182 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=64.216 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=63.062 ms

PC4> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=62 time=35.132 ms

84 bytes from 192.168.5.10 icmp_seq=2 ttl=62 time=43.135 ms

84 bytes from 192.168.5.10 icmp_seq=3 ttl=62 time=43.195 ms

84 bytes from 192.168.5.10 icmp_seq=4 ttl=62 time=36.267 ms

84 bytes from 192.168.5.10 icmp_seq=5 ttl=62 time=33.088 ms

Ping all other PCs from PC5**PC5> ping 192.168.2.10**

84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=81.933 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=81.038 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=84.204 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=80.004 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=86.881 ms

PC5> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=60 time=80.041 ms

84 bytes from 192.168.3.10 icmp_seq=2 ttl=60 time=85.770 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=60 time=86.371 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=60 time=85.269 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=60 time=83.940 ms

PC5> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=62 time=34.224 ms

84 bytes from 192.168.4.10 icmp_seq=2 ttl=62 time=42.160 ms

84 bytes from 192.168.4.10 icmp_seq=3 ttl=62 time=42.056 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=62 time=42.008 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=62 time=43.090 ms

Task 5: Configure clear text authentication between R1 & R4 with pass phrase as Cisco123.**R1 Configuration:**

```
R1#config t
R1(config)#int fa1/1
R1(config-if)#ip ospf authentication
R1(config-if)#ip ospf authentication-key Cisco123
R1(config-if)#exit
R1(config)#exit
R1#
```

***Oct 16 21:53:54.903: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet1/1 from FULL to DOWN, Neighbor Down: Dead timer expired**

R4 Configuration:

```
R4#config t
R4(config)#int fa0/0
R4(config-if)#ip ospf authentication
R4(config-if)#ip ospf authentication-key Cisco123
R4(config-if)#exit
R4(config)#exit
R4#
```

***Oct 16 21:54:08.711: %OSPF-5-ADJCHG: Process 4, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done**

✓ Verification & Testing:**R1#sh ip ospf interface fa1/1**

```
FastEthernet1/1 is up, line protocol is up
Internet Address 14.14.14.1/24, Area 0, Attached via Network Statement
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Topology-MTID Cost Disabled Shutdown Topology Name
  0      1    no    no    Base
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 4.4.4.4, Interface address 14.14.14.4
Backup Designated router (ID) 1.1.1.1, Interface address 14.14.14.1
Flush timer for old DR LSA due in 00:00:59
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:05
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 4.4.4.4 (Designated Router)
```

Suppress hello for 0 neighbor(s)

Simple password authentication enabled

R1#

R4#sh ip ospf interface fa0/0

FastEthernet0/0 is up, line protocol is up

Internet Address 14.14.14.4/24, Area 0, Attached via Network Statement

Process ID 4, Router ID 4.4.4.4, Network Type BROADCAST, Cost: 1

Topology-MTID Cost Disabled Shutdown Topology Name

0 1 no no Base

Transmit Delay is 1 sec, State DR, Priority 1

Designated Router (ID) 4.4.4.4, Interface address 14.14.14.4

Backup Designated router (ID) 1.1.1.1, Interface address 14.14.14.1

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

oob-resync timeout 40

Hello due in 00:00:06

Supports Link-local Signaling (LLS)

Cisco NSF helper support enabled

IETF NSF helper support enabled

Index 2/2, flood queue length 0

Next 0x0(0)/0x0(0)

Last flood scan length is 1, maximum is 1

Last flood scan time is 4 msec, maximum is 4 msec

Neighbor Count is 1, Adjacent neighbor count is 1

Adjacent with neighbor 1.1.1.1 (Backup Designated Router)

Suppress hello for 0 neighbor(s)

Simple password authentication enabled

R4#

Task 6: Configure MD5 authentication between R4 & R5 with pass phrase as Disco123

R4 Configuration:

R4#config t

R4(config)#int fa1/1

R4(config-if)#ip ospf authentication message-digest

R4(config-if)#ip ospf message-digest-key 1 md5 Disco123

R4(config-if)#exit

R4(config)#exit

R4#

*Oct 16 22:04:00.047: %OSPF-5-ADJCHG: Process 4, Nbr 5.5.5.5 on FastEthernet1/1 from FULL to DOWN, Neighbor Down: Dead timer expired

R5 Configuration:

R5#config t

R5(config)#int fa0/0

R5(config-if)#ip ospf authentication message-digest

R5(config-if)#ip ospf message-digest-key 1 md5 Disco123

R5(config-if)#exit

```
R5(config)#exit
R5#
```

```
*Oct 16 22:04:15.911: %OSPF-5-ADJCHG: Process 5, Nbr 4.4.4.4 on FastEthernet0/0 from
LOADING to FULL, Loading Done
```

✓ Verification & Testing:

R4#sh ip ospf interface fa1/1

```
FastEthernet1/1 is up, line protocol is up
Internet Address 45.45.45.4/24, Area 0, Attached via Network Statement
Process ID 4, Router ID 4.4.4.4, Network Type BROADCAST, Cost: 1
Topology-MTID Cost Disabled Shutdown Topology Name
  0      1    no      no      Base
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 5.5.5.5, Interface address 45.45.45.5
Backup Designated router (ID) 4.4.4.4, Interface address 45.45.45.4
Flush timer for old DR LSA due in 00:01:53
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:00
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 3/3, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 5.5.5.5 (Designated Router)
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
Youngest key id is 1
```

R4#

R5#sh ip ospf interface fa0/0 | sec Message

```
Message digest authentication enabled
Youngest key id is 1
```

R5#

Note: Key id should be same on both peers (neighbor)

Task 7: Modify the Hello and Dead Interval between R1 & R4

R1 Configuration:

```
R1#config t
R1(config)#int fa1/1
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#exit
R1(config)#exit
```

R1#

***Oct 16 22:08:19.911: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet1/1 from FULL to DOWN, Neighbor Down: Dead timer expired**

R4 Configuration:

R4#config t

R4(config)#int fa0/0

R4(config-if)#ip ospf hello-interval 15

R4(config-if)#ip ospf dead-interval 60

R4(config-if)#exit

R4(config)#exit

R4#

***Oct 16 22:08:19.911: %OSPF-5-ADJCHG: Process 4, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done**

✓ Verification & Testing:

R1#sh ip ospf interface fa1/1

FastEthernet1/1 is up, line protocol is up

Internet Address 14.14.14.1/24, Area 0, Attached via Network Statement

Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1

Topology-MTID Cost Disabled Shutdown Topology Name

0 1 no no Base

Transmit Delay is 1 sec, State BDR, Priority 1

Designated Router (ID) 4.4.4.4, Interface address 14.14.14.4

Backup Designated router (ID) 1.1.1.1, Interface address 14.14.14.1

Timer intervals configured, Hello 15, Dead 60, Wait 60, Retransmit 5

oob-resync timeout 60

Hello due in 00:00:06

Supports Link-local Signaling (LLS)

Cisco NSF helper support enabled

IETF NSF helper support enabled

Index 1/1, flood queue length 0

Next 0x0(0)/0x0(0)

Last flood scan length is 1, maximum is 1

Last flood scan time is 0 msec, maximum is 0 msec

Neighbor Count is 1, Adjacent neighbor count is 1

Adjacent with neighbor 4.4.4.4 (Designated Router)

Suppress hello for 0 neighbor(s)

R1#

R4#sh ip ospf interface fa0/0

FastEthernet0/0 is up, line protocol is up

Internet Address 14.14.14.4/24, Area 0, Attached via Network Statement

Process ID 4, Router ID 4.4.4.4, Network Type BROADCAST, Cost: 1

Topology-MTID Cost Disabled Shutdown Topology Name

0 1 no no Base

Transmit Delay is 1 sec, State DR, Priority 1

Designated Router (ID) 4.4.4.4, Interface address 14.14.14.4

Backup Designated router (ID) 1.1.1.1, Interface address 14.14.14.1

Timer intervals configured, Hello 15, Dead 60, Wait 60, Retransmit 5

```
oob-resync timeout 60
Hello due in 00:00:11
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 1.1.1.1 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
R4#
```

Note: Hello & Dead intervals should be same on both peers (neighbor)

Ratan

Important Commands:

```
sh ip int brief | exclude unassign
sh ip protocols
sh ip ospf interface brief
show ip route ospf
sh ip ospf neighbor
sh ip ospf database
sh ip ospf interface xx
```

Ratan